

Data Recovery and Safety

CHAPTER 1: INTRODUCTION

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Data and Information

INFORMATION

- Knowledge obtained from investigation, study, or instruction
- In a communication system or computer: A signal or character representing data
- Something justifies change in a construct that represents physical or mental experience or another construct
- In computer science:
 - Organized or classified data, which has some meaningful values for the receiver.
 - The processed data on which decisions and actions are based.

DATA

- A collection of discrete values that convey information, describing quantity, quality, fact, statistics, other basic units of meaning, or simply sequences of symbols. [1]
- In computer science: Data (treated as singular, plural, or as a mass noun) is any sequence of one or more symbols.
- Metadata: data that provides information about other data. [2]

[1]: <https://en.wikipedia.org/wiki/Data>

[2]: <https://en.wikipedia.org/wiki/Metadata/>

The Importance of Data and Information

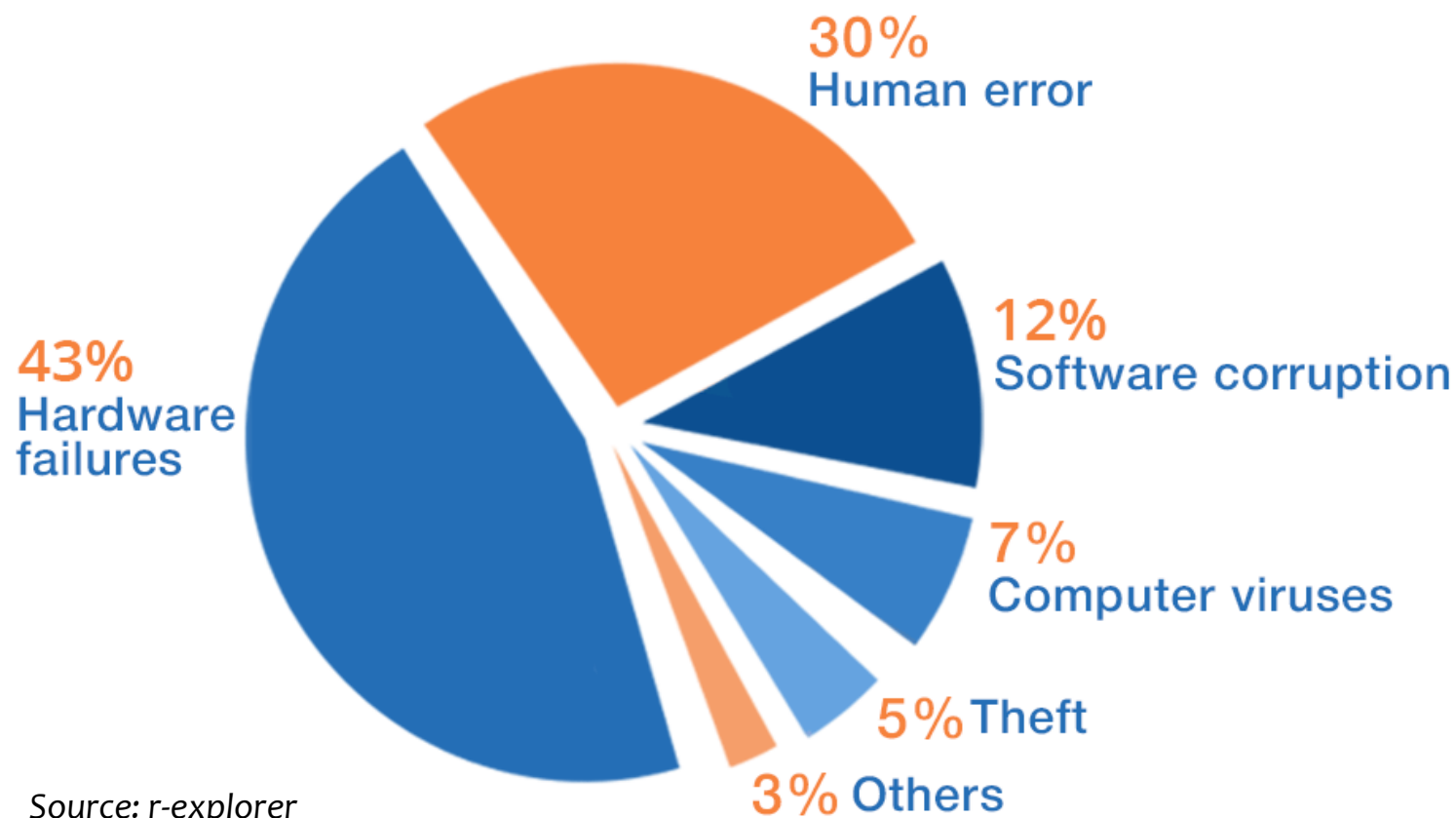
In modern society, information and data are very important. It supports the creation of knowledge, knowledge that helps people avoid mistakes.

In some cases, information and data are even more important than physical assets.

In business, data and information support organizational decision-making and strategy.

Data Loss: Causes, Effects & Prevention Methods

Common Causes of Data Loss



Source: r-explorer

Common Causes of Data Loss

Data Loss

Failure

- ✓ Hardware failures, especially hard drive failure
- ✓ Power failures, resulting in unsaved data to be lost or corruption of files
- ✓ System crash
- ✓ Software corruption
- ✓ File system corruption or database corruption

Unintentional Action

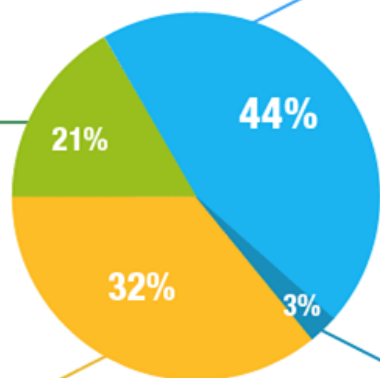
- ✓ Mistaken deletion of files or partitions
- ✓ Misplacement of CDs or Memory sticks
- ✓ Computer viruses attack
- ✓ System hacking

Intentional Action

- ✓ Intentional deletion of files or partitions
- ✓ Hard drive formatting

Disaster

- ✓ Thunderstorms, earthquakes, floods, tornado, fire, etc.



Data Loss Statistics



46% of users have lost data in the past ten years



50% of hard drive die within 5 years



36% of data loss is customer information and financial data



72% of business that suffer major data loss shut down within 24 months



15% of households annually experience theft

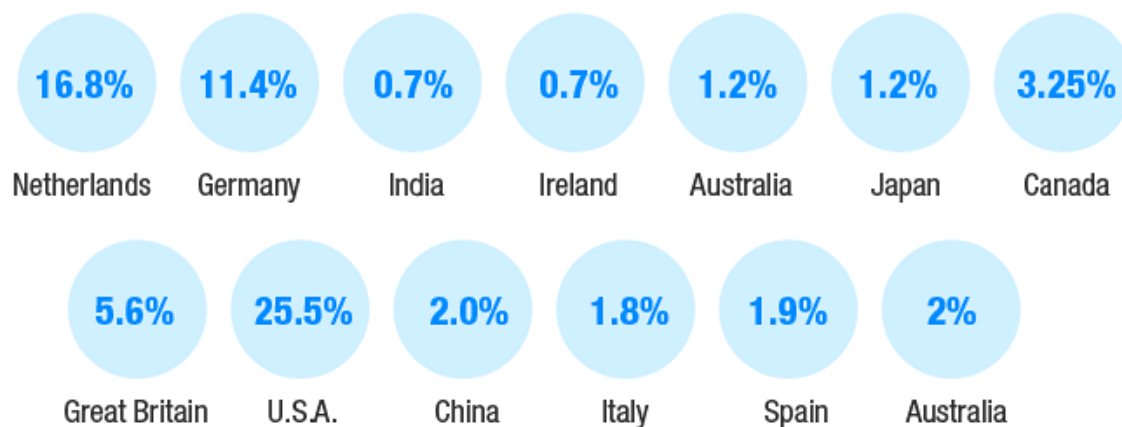
Source: easeUS (2017)

Common Causes of Data Loss

Data Recovery

Lost data actually remains on the storage media, unless it has been overwritten by new data.

Data Loss in Different Countries



Data Recovery Service

- 1.Repair the hard drive
- 2.Image the drive to a new drive
- 3.Logical recovery of files, partitions, and MFT
- 4.Repair the damaged files.

Cost : \$700 - \$1500

Software-based data recovery

- 1.Locate lost files/partitions
- 2.Specify lost file types
- 3.Scan and preview retrieved files.
- 4.Choose the files to recover
5. Save the scanned data for further use.

Cost : \$70 - \$180

Source

<http://mashable.com/2014/04/15/data-breach-infographic/#CCRBkrKeXZqqI>

<http://www.nerdgraph.com/smartphone-data-loss-statistics/>

<http://www.informationsecuritybuzz.com/articles/secrets-data-loss-professionals-wont-tell-infographic/>

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Common Causes of Data Loss

- Data can be lost all of a sudden
- It can happen to anyone, anytime, anywhere, sometimes are predictable and can be avoided, sometimes are not.
- Yahoo have built the security system seriously, however:
 - 2013: 3 billion accounts and personal information were hacked.[1]
 - 2016: 200 million Yahoo! account names and passwords were presented for sale on the darknet market site. [2]
- The trend of sabotage by RansomWare has caused a lot of damage.
 - Between April 2014 and June 2015, the IC3 received 992 CryptoWall-related complaints, with victims reporting losses totaling over \$18 million. [3]

◦ It is estimated that WannaCry caused \$4 billion in damage globally.[4]

[1] <https://www.wsj.com/articles/yahoo-triples-estimate-of-breached-accounts-to-3-billion-1407062804>

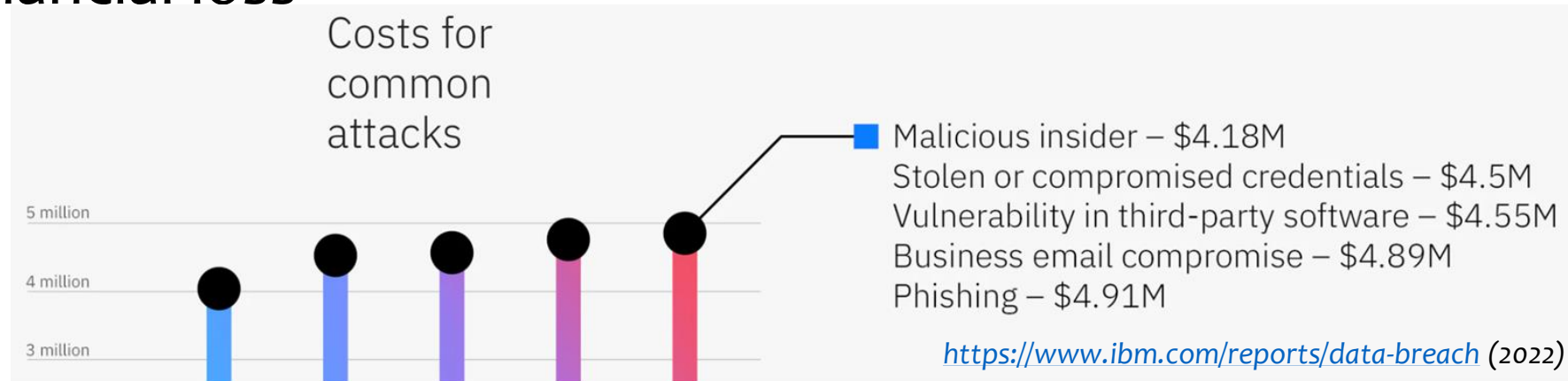
[2] <https://www.vice.com/en/article/aeknw5/yahoo-supposed-data-breach-200-million-credentials-dark-web>

[3] <https://www.ic3.gov/Media/PDF/Y2015/PSA150623.pdf>

[4] <https://www.kaspersky.com/resource-center/threats/ransomware-wannacry>

Effects of Data Loss

- Financial loss



- Broken trust

- o A data breach is a PR disaster that severely damages the reputation of organizations.

- Business closure

Effects of Data Loss

- Disruption
 - Faced with the challenge of data recovery, asking a third party for assistance can expose more information and cause more damage.
 - Data recovery and damage repair is not only a technological challenge, but also disrupts other IT operations of the organization.



<https://www.ibm.com/downloads/cas/L57KW7ND> (2019)

Effects of Data Loss

- Legal action
 - Personal information and user data are always committed to security by organizations.
 - Organizations can face legal penalties for leaking customer data.
 - Ex: 2021: Facebook faces mass legal action over data leak [1]

[1] <https://www.bbc.com/news/technology-56772772>

Effects of Data Loss

Obviously, Data protection and recovery are very important, but:

- “It’s too late to lock the stable when the horse is stolen.” Many people take precautions after data loss has already occurred.
- Conscious people are also often not aware of various situations where incidents can occur and therefore lack preventive measures.
- When the problem of data loss and damage has occurred, how to recover is also a problem that few people know how to solve.

Data protection and recovery capabilities

Data protection capabilities:

- Data can be protected, but many people do not realise or understand this.

Data recovery capabilities:

- In most cases, data can be recovered, but may require professional skills and specialized tools.
- Unfortunately, due to not understanding the mechanisms of data storage & access, users do not see solutions to ensure safety, as well as ways to save data, but also cause serious damage and loss.

Prevent Data Loss

- There are many methods to protect data, including the following essential methods:
 - **Risk Assessments:** Closely guard sensitive data, assess code's trust level before executing.
 - **Backups:** Reducing the risk of data loss. It should be regularly made, updated and stored in a safe place.
 - **Encryption:** Well-encrypted data will be useless and irrecoverable to attackers.
 - **Access Controls:** Access to data needs to be controlled, especially sensitive data. Firewall also needs to be established to control access on the network level.
 - **Destruction:** Discarded data should be destroyed so that it cannot be recovered or accessed by unauthorized parties.
 - **Data Concealment:** Protect confidential data from attackers. Some popular methods:
 - **Obfuscation** makes the message confounding, ambiguous, and harder to understand.
 - **Data Masking** replaces secret information with a non-sensitive version.
 - **Steganography** hides data in another file such as a graphic, audio, or other text file.

Prevent Data Loss

Different types of data recovery methods:

- Hardware processing:
 - When the data storage has a physical failure.
 - Usually need to know about the device structure and operating mechanism.
 - There are still many cases that need little knowledge & experience, but few people know!
- Software processing:
 - When the data storage is not physically damaged.
 - Can use the available data recovery software (Should be done on the disk image)
 - There are many cases where it is not possible to use available software to save, when this requires a lot of knowledge about data organization in the computer.

Prevent Data Loss

Access data directly:

- Winhex:
 - Notice the difference between Logical Volumes and Physical Media.
 - Care must be taken when changing the sector content, as it may cause the operating system to no longer "understand" the data.
 - In case you want to recover data, to be safe, operate on the image volume.
- Programing:
 - BIOS level interrupt functions.
 - Operating system level interrupt functions
 - Operating system API functions
 - Visual C/C# Support

Experiments and exercises

Experiments

1. Crashing your hard disk and recovering it by using WinHex
2. Protecting your data by encryption (MS Office, WinRar)
3. Setting up strong passwords
4. Connecting External Storage Devices
5. Recovering Data by using Data Recovery tool

Exercises

1. Investigate 5 situations of data loss and damage, stating ways to prevent/remediate them.
2. Practice recovering deleted files with Data Recovery Tools.
3. State the possible harms when revealing ID Card
4. Survey and compare at least 3 Data Recovery Tools in WinPE rescue disk
5. Write a program to encrypt the file contents and decode the corresponding program
6. Write a program to lock & unlock Volume by encrypting/decrypting part of the system area.

Open topics

1. Research and presentation on Winhex (focus on user manual)
2. Research and presentation on sector access programming (on an environment different from the one introduced)
3. Identify problems that may arise when programming sector access and how to solve them
4. Research and present a specific data research method.
5. ...

Appendix A

Example of Symmetric cryptography

```
1 #include <stdio.h>
2 #include <string.h>
3 void Encrypt (char * text, int len, char * key) {
4     int keylen = strlen(key);
5     for (int i=0,k=0; i<len; i++,k++) {
6         if (k==keylen) k=0;
7         text[i] ^= key[k];
8     }
9 }
10 void Decrypt (char * text, int len, char * key) {
11     Encrypt(text, len, key);
12 }
13 void main() {
14     char s[] = " * Truong DHKHTN - DHQG TPHCM * ", key[] = "_Khoa_";
15
16     printf("Plaintext: %s \n", s);
17     int len = strlen(s);
18     Encrypt(s, len, key); // ma hoa ban ro s
19     printf("Encrypted text: %s \n", s);
20     Decrypt(s, len, key); // giai ma ban ma s
21     printf("Decrypted text : %s \n", s);
22 }
```

input

```
Plaintext:  * Truong DHKHTN - DHQG TPHCM *
Encrypted text: aH;*0%0%<!Ar >&
               k
Decrypted text :  * Truong DHKHTN - DHQG TPHCM *

...Program finished with exit code 0
```

Appendix A

Encrypt data in any file

```
#include <iostream>
#include <fstream>
#include <string.h>
#define InFileName "d:\\temp\\Document.org"
#define OutFileName "d:\\temp\\x.y"
using namespace std;
void Encrypt(char *text, int len, char * key) { ... }
void Decrypt(char *text, int len, char * key) { ... }
int main(){
    ifstream f_in(InFileName, ios::ate | ios::binary); // open file for input
    if (f_in.fail()) return -1;
    int fileSize = f_in.tellg();
    f_in.seekg(0, ios::beg);
    char * content = new char[fileSize];
    if (!content) {f_in.close(); return -2;} // too simple, need better solution
    f_in.read((char*)content, fileSize*sizeof(int)); // read all file content
    f_in.close();
    Decrypt(content, fileSize, "This is my key"); // (*_*) //
    ofstream f_out(OutFileName, ios::binary); // open Binary file for output
    if (f_out.fail()) return -3;
    f_out.write((char*)content, fileSize);
    f_out.close();
    return 0;
}
```

Appendix A

Exercise 3: It can cause many harms, because it is easy to expose a series of data containing a lot of information that criminals can exploit (tax code, bank account number, address, phone number, date of birth, etc.). This can lead to consequences, such as:

- Revealing household registration information, phone numbers can be bothered by a series of spam ads, or harassed at a heavier level, maybe even being stolen, robbed, harmed...
- Being intimidated by criminals posing as public employees and appropriating property or exploiting confidential/sensitive information and then controlling. (In Vietnam, it is very common to have a fake police's investigation call to scare and appropriate a lot of money).
- The ID can register phone subscribers, make credit cards or loan documents with the name and address of the person exposed. Fraudsters receive money and those whose information is leaked must pay or are greatly disturbed by the lender's debt collection.
- ID information can forge documents, contracts, and perform illegal transactions. Until being investigated, held accountable, claimed for compensation, etc., then we know and sometimes we can't prove that we are innocent!
- From the ID information, sometimes criminals have stolen phone numbers and bank accounts at a level where the exposed person didn't even know that they were robbed and when they knew it, it was too late.
- The information on the ID can register the establishment of fraudulent companies (in the director's name). After causing much harm, the director is the person accused.

Questions and Answers